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University of the Punjab, Lahore

Smart Academic Communication Platform (SACP)

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Statement of Submission

In this regard, we formally declare that the project titled "**Smart Academic Communication Platform**", which has been done in fulfillment of the requirements for the **Bachelor of Science in Computer Science (BSCS)** degree at **Faculty of Computing & Information Technology, University of the Punjab, Lahore**, is our original effort.

This project has been conducted under the supervision of the project supervisor assigned to us. It is hereby declared that this project has never been submitted, wholly or partially, for obtaining any degree or diploma from any other university or institution before.

We have properly mentioned all the references used in this project.

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Special Thanks

- Person A
- Person B

Date: []

Abstract

Academic institutions often encounter difficulties in organizing effective communication between administrators, faculty members, and students. Traditional communication systems are manual, inefficient, delay-prone, and hard to scale. Important notifications, such as warnings about attendance, assignment deadlines, test schedules, and institutional communications, may be late and not delivered to their respective recipients on time. This reduces the effectiveness of the communication process and causes organizational problems.

Smart Academic Communication Platform is an integrated, intelligence, automated, and web-based tool to facilitate efficient academic communications. It offers rule-based automation of institutional communication processes and personalization of messages based on institutional data to create context-aware and personalized messages. Such features include dynamic messages templates, campaigns scheduling, target audience selection, notification tracking, and analytics.

The architecture of the project relies on a distributed 3-tier structure that utilizes React for front-end applications, ASP.NET core backend services, Node.JS-based real-time communications, and AI implemented services using Python. Machine learning algorithms can help to classify and prioritize emails, thus preventing spam issues and focusing more on relevant academic messages. Other functionalities offered by SACP include real-time chat options, communication logging tools, reports, and documents sharing.

The main purpose of the project is to provide institutional communication solutions with automation capabilities and improved message delivery, which allows for reduced manual efforts and intelligent assistance with communication tasks.

The expected result of the project development is a functional academic communication platform that provides scalable, usable, centralized management and analytics capabilities.

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Chapter 1

INTRODUCTION

1.1 Background

Communication plays a critical role in ensuring that higher education institutions run smoothly. There are constant interactions between administrators, lecturers, and learners with regard to lectures, exams, assignment deadlines, scheduling, notifications, and other academics. Most universities and colleges use conventional communication means such as emails, social media groups, printed notices, or external messaging apps in their interactions. The problem with these communication strategies is that they lead to ineffective messaging, poor message management, lack of personalization, and difficult large-scale communications.

In light of the current digitized world and data-driven systems, institutions need communication platforms which can automatically streamline their communication procedures. There are existing solutions which are learning management-oriented or notification-oriented but they lack communication management, automation of communication workflows, analytics, intelligent prioritization, and targeting features.

Smart Academic Communication Platform (SACP) will provide an innovative, central, and intelligent communication solution specifically designed to serve the needs of institutions. Automation will be achieved using rule-based workflows, campaigns, dynamic templates, real-time messaging, and email classification and prioritization using artificial intelligence technology. Integration with academic records will ensure that there is personalized and context-aware communication among administrators, educators, and learners.

Distributed architecture principles will also be addressed through the development of specific layers for frontend, backend core services, real-time communication services, and artificial intelligence services. It will provide an automated, analytical, and AI-powered system which aims to streamline academic institution communication procedures.

1.2 Problem Statement

In academic institutions, many communication issues arise due to the use of manual and semi-manual mechanisms. Academic notifications like warnings about poor attendance, deadlines for assignments, exam schedules, fee collection reminders, and other important announcements are often made through inefficient or inconsistent communication processes. The current system of communication is ineffective because it does not offer personalization, automation, centralization, and intelligent message priority.

In addition, organizations operating thousands of students and faculty face the following challenges:

- Sending targeted and personalized communications
- Managing large communication campaigns
- Monitoring deliverability and engagement metrics
- Filtering out spam and irrelevant content
- Automating repetitive communication processes
- Centralizing communications record-keeping and analysis

- Securing role-based communication lines

Manual systems fail to provide suitable automation capabilities and real-time collaboration opportunities. With the increase in communication volume, organizations need scalable technology solutions to automate and optimize their communication processes.

There is thus a need for a centralized, automated, and intelligence-based system to support personalized communications, workflow automation, communication analytics, real-time communications, and machine-learning-based message priorities.

1.3 Objectives

The main objective of the SACP solution is to create a smart communication platform for academic institutions, which will automate the process of communication and increase its efficiency.

The key objectives of the project include the following:

- Create a centralized platform for managing academic communication.
- Utilize dynamic templates to support personalized communication.
- Introduce automated and rule-based workflows to be activated as a result of certain academic events, such as missed class time, due dates for assignments, exam dates, etc.
- Manage campaigns and send bulk emails/messages.
- Allow for targeted communication according to academic data.
- Keep track of emails via logs and delivery reports.
- Develop analytics dashboards to monitor communication performance.
- Apply machine learning techniques to categorize and prioritize emails to ensure that only important emails are delivered.
- Introduce safe and instant communication capabilities within the platform.
- Decrease the manual work required to send emails and increase efficiency.

1.4 Scope of the Project

The scope of the Smart Academic Communication Platform (SACP) includes design and implementation of the web-based, distributed communication platform specifically for academic institutions. The goal of the platform is to improve the institutional communication capabilities through automation, personalization, analytics, and smart communication management.

The key functional domains of the platform include:

- User authentication & role-based authorization
- Management of academic records
- Template management (dynamic templates)

- Campaign creation & scheduling
- Rule based workflow automation
- Email & notifications service
- Real-time one-to-one communication
- AI-powered email classification/prioritization
- Communication analytics & reporting
- Document/file exchange

Key target audiences of the SACP platform are the following:

- Academic Administrators
- Faculty members
- Students

Regarding architecture of the application, the platform follows the distributed approach, where frontend services, backend/core services, real-time communications services, and AI-based services are independently developed to increase maintainability and scale of the platform.

Among others, exclusion domains of the project include the following:

- Payment and financial operations
- Complete functionality of LMS platform
- Social messenger type applications like WhatsApp, Messenger

It is worth noting that the core of the project does not include any course management functions.

1.5 Methodology Summary

The methodology of development on which the SACP has been developed uses an iterative approach for software development. The software engineering methodology also includes the integration of other concepts such as the concept of distributed architecture, databases, and use of machine learning techniques.

The software development process includes the following stages:

1.5.1 Requirement Analysis

We analyze communication related problems within academic institutions in this phase. Based on the needs of academic institutions functional and non-functional requirements are identified.

1.5.2 System Design

In this phase, we design the entire system of SACP including system architecture, database design, workflow models, communication modules, automation engine, and AI integration components are designed using UML diagrams, ERDs, and architectural models

1.5.3 Frontend Development

In this phase, we design and develop user interfaces using React and modern UI libraries to provide user-friendly and responsive interfaces for administrators, faculty members, and students.

1.5.4 Backend Development

In this phase, all services related to backend core are implemented using ASP.NET Core Web APIs. These services handle authentication & authorization, campaign & template management, automation workflows, analytics and AI orchestration with Python services.

We'll also develop real-time backend services for communication module and real time analytics for dashboard using SignalR and Redis within .NET ecosystem.

1.5.5 AI/ML Service Development

In this phase, we'll develop AI/ML services for email classification and prioritization using Python. We'll develop, finetune, or create machine learning models and NLP techniques to achieve this.

1.5.6 Database Integration

In this phase, we'll setup either PostgreSQL or Azure SQL database to store academic records, communication logs, templates, campaigns and analytics data. We'll use a centralized database to avoid duplication, redundancy, and confusions.

1.5.7 Testing and Validation

In this phase, we'll conduct unit testing, integration testing, API testing, AI model evaluation, security testing, and performance testing techniques to ensure overall reliability and functionality of the SACP.

1.5.8 Deployment and Evaluation

In this phase, we'll deploy the final system in a distributed environment considering scalability, usability, and communication performance among system components.

1.6 Tools and Technologies Used

The Smart Academic Communication Platform (SACP) is developed with modern web, backend, database and AI technologies to provide scalability, maintainability and performance.

1.6.1 Frontend Technologies

- React
- HTML5
- CSS3
- JavaScript
- Tailwind CSS / Material UI
- Redux
- Charts.js / Recharts

1.6.2 Backend Technologies

- C#
- ASP.NET Core Web APIs
- Entity Framework Core
- LINQ
- JWT Authentication
- SignalR
- Redis
- Hangfire / Quartz.NET

1.6.3 AI/ML Services

- Python
- FastAPI / Flask
- Scikit-Learn
- Pandas
- NumPy
- spaCy / NLTK
- Matplotlib

1.6.4 Database System

- PostgreSQL / Azure SQL

1.6.5 Development and Collaboration Tools

- Visual Studio
- Visual Studio Code
- PyCharm
- Git & GitHub
- Swagger / Postman
- Draw.io
- PlantUML
- Figma
- Jira / Trello
- MS Office

1.6.7 Deployment Environment

- Linux
- Cloud-based deployment support
- Distributed service architecture

1.7 Summary

This chapter introduced the Smart Academic Communication Platform (SACP), its background, aims, scope, methodology and technologies used. The chapter explored the communication challenges that academic institutions face and explained how automation, distributed architecture, and machine learning can help improve communication workflows. The next chapter includes the literature review and analysis of existing systems related to academic communication, notification management, collaboration platforms, and AI-based communication solutions.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

Academic institutions mostly rely on digital communication platforms to manage their interactions among faculty members and students. A simple, effective, and autonomous communication is essential for delivering timely messages, but still many institutions use tools which are made for general public and specialized for their specific use cases like tracking engagement, sending messages to large audience on specific time or events, etc.

There are many modern platforms, such as Slack, Microsoft Teams, Discord, etc., which provide real-time communication and collaboration. But email classification and campaign systems, template systems, and automated workflows for specific needs of academic institutions are missing. Current solutions typically focus on isolated communication tasks and rarely offer a unified academic communication system for their specific needs.

The proposed Smart Academic Communication Platform (SACP) seeks to overcome these challenges by integrating all necessary features very specific to academic institutions into a single distributed system architecture.

This chapter reviews existing communication platforms, notification systems, automation tools, and intelligent messaging solutions related to SACP.

2.2 Summary of Existing Systems and Research

2.2.1 Realtime Communication Platforms for Education

2.2.1.1 Slack

Slack is a cloud-based collaboration tool used in corporate and educational settings for real-time communication, group interaction, and file sharing.

Feature Category	*Slack Capabilities*	*What Slack Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time & threaded messaging, dedicated classroom/announcement channels, and audio/video meetings & file sharing	It lacks large-scale SMS/email broadcasting and automated academic notifications	It offers rule-based academic automation and workflows for personalized notifications
Integrations	It integrates with 3rd-party apps & productivity tools like LMS integration (Canvas/Blackboard) via LTI Course Bots	It lacks centralized campaign management	It offers scheduled communication campaigns and unified location for multiple academic record sources
Intelligence & Analytics	It offers basic workspace synchronization	It lacks mL-based email classification and advanced	It offers AI-based message prioritization and comprehensive

		academic analytics	communication & workflow dashboard

While Slack excels at daily, real-time classroom collaboration, it lacks institutional management tools. The proposed SACP expands on Slack by adding AI prioritization, advanced analytics, automated workflows, and centralized academic record integration.

2.2.1.2 Microsoft Teams

Microsoft Teams is an enterprise-wide communication platform deeply integrated with Microsoft 365 Education, focusing on daily classroom collaboration, assignments, and video meetings.

Feature Category	*Microsoft Teams Capabilities*	*What Teams Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time chat & class channels, video conferencing & calendar sync, and native assignment/classwork management	It lacks native external messaging (SMS/WhatsApp), and multi-layered distributed structure	It offers multi-layer distributed architecture for external messaging, and optional multi-channel outreach
Integrations	It offers deep Microsoft Office app ecosystem, and LMS sync & School Data Sync (SDS) for grades	It lacks flexible, institutional campaign management tools	It offers centralized communication campaigns, and academic event-driven automation
Intelligence & Analytics	It offers basic classwork activity tracking	It lacks AI-driven priority assignment, and advanced notification analytics	It offers intelligent email filtering, and analytics-driven communication workflows

Microsoft Teams is built primarily for real-time collaboration and classroom management. The proposed SACP shifts the focus from simple collaboration to high-level institutional automation, offering AI-driven filtering, advanced workflow analytics, and event-driven automation engines.

2.2.1.3 Discord

Discord is a community-centric platform originally built for gaming that has evolved into an informal educational tool used for real-time virtual classrooms, voice chat, and student-run communities.

Feature Category	*Discord Capabilities*	*What Discord Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers real-time text, voice, & video chat, community-based	It lacks out-of-platform notifications, and	It offers centralized academic communications tracking,

	servers & channels, and screen sharing & basic file sharing	centralized communication tracking	and AI-driven communications prioritization
Integrations	It offers multi-platform accessibility, and basic announcement channels	It lacks native LMS integrations, and centralized academic record syncing	It offers seamless integration of academic records, and workflow automation built for institutions
Administration	If offers informal server roles and basic text permissions	It lacks proper institutional user structuring, and academic analytics & reporting	It offers role-based academic communication management, and admin dashboard with academic & operational analytics

Discord is specialized at creating highly engaging, real-time community spaces, but it completely lacks enterprise-level administration, data compliance, and tracking. The proposed SACP bridges this gap by introducing rigorous role management, institutional workflow automation, and real-time administrative dashboards.

2.2.1.4 Rocket.CHAT

Rocket.Chat is an open-source, self-hosted communication platform favored by privacy-conscious institutions for its strict data security control, flexibility, and real-time messaging infrastructure.

Feature Category	***Rocket.Chat Capabilities***	***What Rocket.Chat Lacks***	***Proposed SACP Enhancements***
Core Communication	It offers real-time chat & channel collaboration, secure, private messaging, and direct file sharing capabilities	It lacks academic workflow tools, and centralized, campus-wide logging	It offers built-in academic communication workflows, and centralized communication logging
Integrations	It offers flexible third-party application integrations	If lacks development effort required to adapt for complex academic logic	It offers academic event-triggered automation, and native, plug-and-play institutional triggers
Security & Analytics	It offers on-premises, self-hosted data control, and compliance-ready secure environments	It lacks intelligent message classification or filtering, and dedicated institutional analytics dashboards	It offers AI-based communication classification, and comprehensive analytics and reporting modules

Rocket.Chat provides an incredibly secure and flexible canvas for communication, but it requires massive development overhead to build out institutional features. The proposed

SACP eliminates this friction by delivering pre-built academic automation engines, AI classification, and deep data analytics.

2.2.2 Email Campaign and Communication Automation Platforms

2.2.2.1 MailChimp

MailChimp is a cloud-available marketing automation and email service to support large scale campaigns through the ability to segment audience and track performance as part of a general commercial approach to marketing.

Feature Category	*MailChimp Capabilities*	*What MailChimp Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers bulk campaign creation & scheduling, drag-and-drop template libraries, and dynamic placeholder fields	It lacks native multi-channel academic networks, and institutional role mapping	It offers campaign management module, and template management system natively with academic role distribution.
Audience & Automation	It offers audience segmentation based on lists, and standard user event-triggered emails	It lacks native integration with academic registers, and event engines built for school systems	It natively offers academic data management, and automation rule engine to make communication automated and context-aware with engagement analytics

MailChimp provides excellent tools for managing commercial campaigns and templates, but there are very tightly controlled manual processes required to protect sensitive student data from becoming exposed. The proposed SACP will replicate the same marketing functionality, but will do so natively in institutional data model(s), provide depth of analytical capabilities, and have inherent educational privacy standards built into the system.

2.2.2.2 SendGrid (Twillio SendGrid)

Sending transactional emails through Twilio SendGrid is simple, scalable, and hassle-free; sending email is done through the API with no need to manage infrastructure, while providing reliable transactional emails to your recipients.

Feature Category	*SendGrid Capabilities*	*What SendGrid Lacks*	*Proposed SACP Enhancements*
Messaging Infrastructure	It offers high-volume transactional email delivery, and API-driven framework optimized for heavy messaging loads	It lacks built-in application UI for non-technical academic staff	It offers core messaging service with ASP.NET Core & Hangfire backend optimizations

Tracking & Logging	It offers real-time processing receipts, and detailed delivery logs & bounce-rate tracking	It lacks academic context tracking (linking logs directly to student/faculty profiles)	It offers detailed communication logs, and centralized student-to-message history mapping
Architecture & Performance	It contains heavy infrastructure designed for offloading primary app threads	It lacks pre-built educational logic, automation rules, or template engines	It contains Hangfire-driven background processing integration, and native academic campaign & rule engines

Using the SendGrid email infrastructure as a "raw" email delivery and tracking mechanism, the SACP delivers high-performance background processing (using Hangfire) along with a complete UI, data models and automated workflows that customers need to run their educational institution.

2.2.2.3 Braze

Braze is a cutting-edge customer engagement platform with an enterprise-level architecture based on high-velocity event stream processing, real time event triggers and complex cross channel user journeys.

Feature Category	*Braze Capabilities*	*What Braze Lacks*	*Proposed SACP Enhancements*
Automation & Journeys	It offers real-time lifecycle behavior triggers, and visual cross-channel journey building (Canvas Flow)	It lacks native academic-specific entity triggers (e.g., GPA drops, prerequisite failures)	It offers automation rule engine, and pre-configured institutional event templates
Data & Integrations	It offers streaming data ingestion from warehouses, and dynamic content personalization via external APIs	It lacks integrations with standard school registers and educational records	It offers academic data management to store all academic records
Intelligence & Analytics	It offers real-time user interaction tracking, and predictive churn and behavioral AI modeling	It lacks contextual academic classification (e.g., urgent administrative alerts vs. student club notifications)	It offers analytics and reporting. and AI-based academic communication classification

Braze delivers superior levels of real time event processing and automated behavior, however to adapt it to work in a higher education environment would require a considerable amount of data mapping and custom development effort by the academic institution utilizing the Braze platform. The SACP will duplicate the same level of real time agility and visual workflow functionality as provided by Braze, but will do so natively as part of the school's record system(s) and academic classification system(s).

2.2.3 Academic Notification and Student Information Systems

2.2.3.1 Canvas LMS

Canvas LMS has an extremely powerful Student Notifications hub focused on delivering active course content; and providing assignment due dates and role-based messaging to learners directly. Through its use as a local course-level alert, the LMS can effectively be used to provide information at the local level. But when it comes to institutional outreach and automating processes related to communication and recordkeeping, the LMS cannot be used in conjunction with the institution's operations.

Feature Category	***Canvas LMS Capabilities***	***What Canvas Lacks***	***Proposed SACP Enhancements***
Multi-Channel Delivery	It offers email (individual & daily digests), mobile push notifications & in-app activity streams, and user-level frequency preferences	It lacks real-time 1-to-1 chat networks, and direct external channel fallbacks (SMS/WhatsApp)	It offers real-time 1-to-1 conversational chat, and backend-driven multi-channel notification architecture
Audience & Campaigns	It offers "Message Students Who" data-filters, and global admin banner broadcasting by enrollment role	It lacks centralized campaign scheduling interfaces, and dynamic, reusable message template systems	It offers campaign Scheduling & Bulk Communication, and dynamic message templates with record variables
Automation & Analytics	It offers immediate or batched course event notifications	It lacks event-driven trigger engines (e.g., attendance drops), open/delivery tracking & communication audits, and intelligent message sorting or prioritization	It offers rule-based workflow automation engine, delivery tracking & meticulous communication logs, analytics & engagement reporting dashboard, and AI/ML email classification & prioritization

This proposed SACP will serve as the primary high-level intelligence layer for schools and will offer school administrators the ability to perform rule-driven automation; gain access to advanced analytics related to student engagement with courses and use machine learning to filter out institutional noise.

2.2.3.2 Moodle

Moodle is an open source and as such has numerous plugins for additional capabilities; however, its flexibility means it can serve multiple purposes, including course level communication and chat.

Feature Category	*Moodle LMS Capabilities*	*What Moodle Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers native direct 1-to-1 & group chat, in-platform web alerts & bulk messaging, and mobile push via AirNotifier (APNs/FCM)	It lacks automation, and direct commercial multi-channel fallback	It offers built-in academic event-triggered alerts
Campaigns & Data	It offers simple bulk student messaging by course, and custom message API for plugins	It lacks centralized campaign scheduling interfaces, and reusable template engines with record data	It offers campaign scheduling & management UI, dynamic message templates with record variables
Automation & Analytics	It offers per-user delivery channel preferences, and administrator institution-wide defaults	It lacks complex, rule-based academic trigger engines, open, read, and bounce-rate tracking, and intelligent message triage and sorting	It offers rule-based workflow automation engine, delivery Tracking & Analytics dashboard, and AI/ML Email Classification & Prioritization

Moodle provides an excellent, adaptable open-source foundation for course-level communication and chat, but extending it to handle campus-wide operational automation requires heavy plugin reliance and customization. The proposed SACP focuses on eliminating this fragmentation by providing robust campaign management, rules-based workflow engines, full delivery analysis and AI filtering as part of the core platform architecture.

2.2.4 SMS and Messaging Platforms

2.2.4.1 TrueDialog

TrueDialog is a classy SMS/MMS and WhatsApp communication delivery application designed specifically to support large scale communications at high volumes while obeying to all Carrier compliance requirements while allowing visibility into message volumes and engagement metrics.

Feature Category	*TrueDialog Capabilities*	*What TrueDialog Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers high-volume SMS, MMS, & WhatsApp outreach, core delivery tracking & logging metrics, and	It lacks native, real-time 1-to-1 in-app chat networks, and centralized & cross-channel email routing	It offers real-time conversational chat module, and unified email & SMS fallback infrastructure

	text-based contact engagement		
Integrations & Data	It offers commercial CRM and database integrations, and consent and opt-out management systems	It lacks native context-awareness regarding academic profiles and course structures	It offers direct integration with institutional academic records, and context-aware data mapping
Automation & Intelligence	It offers broadcast scheduling and basic text triggers	It lacks rules driven by academic events (e.g., GPA drops, attendance, exam dates), and ML-driven priority sorting and classification	If offers academic event-triggered automation engine, AI/ML message classification & prioritization, and comprehensive operational analytics dashboards

The proposed SACP goes much farther than TrueDialog in that it utilizes the same communication channels (Text messaging/SMS and WhatsApp) to establish an academic-first framework so that these channels of communication are fused with intelligent machine learning for triage, real-time synchronization of records and automation rules for institutions.

2.2.4.2 WhatsApp Business

WhatsApp's Business messaging channel is extremely popular all over the globe with incredible open rates, the ability to send rich content, a secure encrypted connection between sender/receiver, and can send messages with interactive buttons.

Feature Category	***WhatsApp Business Capabilities***	***What WhatsApp Lacks***	***Proposed SACP Enhancements***
Core Communication	It offers instant, high-engagement text & media delivery, interactive messaging components, and end-to-end encrypted delivery channels	It lacks unified, cross-platform email fallbacks, and flexible, unvetted outbound text broadcasting	It offers centralized multi-channel management layer, and unified email, SMS, and messaging orchestration
Governance & Data	It offers strict opt-in handling & privacy protection, and commercial template pre-approval systems	It lacks direct out-of-the-box integration with student information systems	It offers direct integration with institutional academic records, and centralized academic communication governance

Automation & Intelligence	It offers basic interactive messaging automated replies	It lacks triggers tied to real-time school events, advanced notification performance reporting, and ML-based message triage and prioritization	It offers academic event-triggered automation engine, AI/ML message classification & prioritization, and built-in campaign management & delivery analytics

The WhatsApp Business channel has a vast conversational engagement and exit rates but only functions as an independent messaging service and must rely on an external system to attach data and controls to a message or workflow; this solution to this disconnect is the SACP in that it serves as an intelligence layer to control all outbound messaging channels including sync with internal school roster/s and delivers messages according to pre-built academic based rules and/or machine learning priority system.

2.2.5 AI-Enhanced Email and Notification Tools

2.2.5.1 Spark Mail

The Spark Mail app's functionality includes automated summarization, intelligent filtering of your inbox, and AI writing assistant features to combat information overload and boost your productivity as an individual or as part of a team.

Feature Category	***Spark Mail Capabilities***	***What Spark Mail Lacks***	***Proposed SACP Enhancements***
Core Communication	It offers smart Inbox filtering (Personal, Notifications, News), real-time team email sharing & commenting, and unified client for standard email ecosystems	It lacks centralized multi-channel broadcasting (SMS/WhatsApp), and native cross-platform conversational chat layers	It offers centralized multi-channel orchestration layer, and ntegrated real-time conversational chat module
AI Capabilities	It offers Spark +AI writing & drafting assistance, and automated thread summarization & quick replies, and basic priority sorting for standard emails	It lacks classification based on academic context (e.g., grading alerts vs. campus events)	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers basic personal email templates and snoozing	It lacks triggers tied to external databases or school events, campus-wide delivery tracking and audit logs, and centralized	It offers academic event-triggered automation engine, campaign administration module, and

		administrator monitoring dashboards	institutional analytics & reporting

Although it's a great end user application for personal mail management and mail drafting, Spark Mail is actually just a client over existing mail account. The proposed SACP (Student Account Provisioning Component) will be an entire enterprise-grade infrastructure using similar AI-driven triage functionality and built on a multi-layer distributed architecture (front-end, back-end, real-time, and AI components) that interfaces with existing school record systems to provide automated, institution-wide workflows.

2.2.5.2 SaneBox

Using cloud technology to organize, prioritize and track emails is what SaneBox is all about. By connecting to your email server through IMAP and utilizing the metadata associated with your incoming email messages, SaneBox can filter out many of the irrelevant messages that come into your inbox and only provide you with the messages that you are interested in.

Feature Category	***SaneBox Capabilities***	***What SaneBox Lacks***	***Proposed SACP Enhancements***
Core Communication	It offers smart filtering folders (@SaneLater, @SaneNews), one-click sender banishing via SaneBlackHole, and works invisibly across any existing email client	It lacks native multi-channel, broadcasting (SMS/WhatsApp), and in-app conversational 1-to-1 chat environments	It offers centralized multi-channel broadcasting engine, and integrated real-time conversational chat module
AI Capabilities	It offers server-level header & metadata pattern tracking, and contextual reply generation via SaneDrafts	It lacks deep text body NLP analysis, and classification based on academic-specific context	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers response loop monitoring via SaneNoReplies, and time-based follow-up nudges (SaneReminders)	It lacks triggers tied to external databases or school events, and campus-wide delivery tracking and audit logs	It offers academic event-triggered automation engine, campaign administration module, and institutional analytics & reporting

By utilizing machine learning and artificial intelligence, SaneBox creates an incredibly simple method to filter and track all of the incoming email that you receive. The SACP is designed to take that filtering process and expand it out to an entire organization by embedding the artificial intelligence filtering process into a complex distributed architecture and leveraging explicit education record metadata to create automated campus wide processes.

2.2.5.3 Shortwave

Shortwave is an advanced AI-native email client engineered around Large Language Models (LLMs) and the open Model Context Protocol (MCP), specializing in natural-language inbox triaging, semantic search, thread summarization, and cross-app workflow connections.

Feature Category	*Shortwave Capabilities*	*What Shortwave Lacks*	*Proposed SACP Enhancements*
Core Communication	It offers task-focused inbox bundles & splits, real-time shared team email threads & private sidebar comments, and native delivery scheduling & granular push settings	It lacks strict cross-platform SMS or WhatsApp orchestration fallback, and dedicated, secure campus-wide data routing	It offers multi-channel broadcasting, and integrated real-time conversational chat module
AI Capabilities	It offers automatic "TL;DR" email & attachment summaries, generative drafts mirroring unique writing styles, and custom plain-English AI inbox filtering scripts	It lacks categorization mapping based strictly on institutional schemas (e.g., academic status vs. course registration)	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers MCP & Zapier loops to trigger 3,000+ external tools (Notion, Asana), and instant interactive scheduling & calendar generation	It lacks direct hooks into student information records, and global institutional audit logs and administrative oversight	It offers academic event-triggered automation engine, campaign administration module, and institutional analytics & reporting

Shortwave showcases how powerful LLM integrations can fundamentally rewrite email productivity, execution tracking, and app workflows for individual professionals or teams. The proposed SACP scales these precise conversational and processing concepts up to the structural level—embedding intelligence directly into the institution's core data models to govern, automate, and analyze campus-wide communications automatically.

2.2.5.4 Microsoft Outlook Copilot

Microsoft Outlook AI (powered by Microsoft Copilot and Azure OpenAI) is a mature, enterprise-grade email and productivity engine designed to optimize personal inbox management, draft communication, and synthesize calendar workflows.

Feature Category	*Outlook AI Capabilities*	*What Outlook AI Lacks*	*Proposed SACP Enhancements*

Core Communication	It offers deep Microsoft 365 ecosystem fluid sync, advanced Mail Merge for mass personalization, and real-time meeting readiness dashboards	It lacks non-email channels (SMS/WhatsApp), and native student conversational chat partitions	It offers multi-channel broadcasting with integrated real-time conversational chat module
AI Capabilities	It offers dynamic thread summaries & writing coaching, user-level *Prioritize My Inbox* logic rules, and text drafting matching unique individual voices	It lacks sorting grounded specifically in institutional data objects (e.g., academic standing, registration holds)	It offers AI-based academic communication classification, and ML-driven institutional prioritization
Automation & Analytics	It offers basic personal follow-up tracking & reminders, and Work IQ semantic organization model filters	It lacks automation loops triggered by live campus data (e.g., low attendance, deadline failures)	It offers academic event-triggered automation engine, campaign administration module, and institutional analytics & reporting

Microsoft Outlook Copilot represents an impressive standard for user-level productivity, thread summarization, and writing assistance within a corporate inbox framework. The proposed SACP scales past personal inbox triage by implementing a distributed multi-layer architecture—fusing AI ranking models with direct student information records to coordinate automated, institution-wide communication policies.

2.3 Analysis of Gaps and Limitations in Current Solutions

In this chapter, we've reviewed several systems that represent a large leap forward in new ways to communicate, work collaboratively and communicate via alert systems, automate workflows, and communicate using AI-based messaging systems. These systems are very strong in their individual components, however, when looking at the potential to build a better academic communication ecosystem from existing systems, it became apparent most current systems provide solutions to narrow areas of communication management and therefore do not support an integrated academic communication ecosystem. Therefore, educational providers are using multiple, non-integrated systems to manage notifications, messaging, campaigns, analytics, and academic communication workflows, leading to inefficiencies and administrative burden.

The literature review identified many key limitations in existing systems.

2.3.1 Lack of Centralized Academic Communication Management

The majority of communication platforms focus on general purpose collaboration, business communication or engagement with customers — certainly not intended as a mechanism

for managing academic communication. The overwhelming majority provide you with the ability to messaging effectively; however, they do not integrate directly with your academic records, reporting systems or the institutional workflow.

The majority of email automation platforms, SMS communication applications all have their primary focus on delivering messages; however, the majority will require external interfaces in order to retrieve student records, attendance information, course data and information related to academic events.

As a result, the majority of higher education institutions end up using multiple, disconnected systems for supporting their communication process, which creates disconnected data and has a negative impact on staff productivity and work efficiency.

2.3.2 Limited Workflow Automation

Most of the systems in use are capable of sending messages and notifications but only support a very limited number of automated communications for creating, managing and executing a variety of workflows in an academic setting.

Automated communication is frequently required within educational institutions for a number of different situations that include:

- Attendance Shortage Warnings
- Deadlines associated with Assignments
- Exam Schedules
- Student Interventions and Retention Procedures
- Notification Chains for faculty members and students.
- Processes for Administrative Approvals
- Personalized Communication Campaigns

Some products such as Shortwave and other workflow automation tools provide limited amounts of automation; however, they were not designed specifically for educational institutions, and there are no methods available for the seamless integration of academic processes with these tools.

The proposed SACP (Smart Academic Communication Platform) would eliminate these limitations by providing a configurable rule-based workflow engine that will produce automated correspondence for an academic institution or user based upon academic data and events within an institution.

2.3.3 Absence of AI-based Classification and Prioritization System

Spark Mail, SaneBox, Shortwave, and Microsoft Outlook Copilot are all new tools used in combination with AI-powered technologies to provide users with additional features such as intelligent inboxes, summaries of emails, intelligent searches & priority types of emails.

These tools are designed primarily to enhance personal productivity & improve enterprise communications; while these tools are helpful for users in these contexts – they do not focus on helping users manage their communications for an educational institution. The way they

determine which types of communications to prioritize, generally uses data surrounding: (Sender Behavior, User Interaction, Key Words, Conversations) without taking academic data into account.

Because of this limitation, users such as students, faculty & administration continue to have an increased number of notifications making it possible to miss: (Deadlines, Announcements, Events).

The proposed Student-Academic Communication Platform (SACP) will utilize machine learning models to assist users with:

- Identifying spam communications
- Identifying academic notifications to prioritize
- Intelligent message categorization
- Identifying high-risk communication scenarios
- Providing context-aware notifications using institutional data

These additional capabilities will help support the communication needs of educational institutions.

2.3.4 Weak Academic Analytics and Reporting

The majority of systems evaluated offer primary communication metrics: message delivery rate, open rate, click-through rate, and engagement statistics.

These metrics, although helpful, but do not provide enough data for evaluating institutional communication strategy effectiveness. Educational institutions need additional reporting capability, including:

- Analysis of overall communication campaign performance
- Monitoring of student engagement
- Visualization of notification delivery
- Measurement of academic communication effectiveness
- Reporting of all departmental communications
- Auditing of all institutional communications

Advanced analytical capabilities in the area of education analytics are critical to enabling the institution to make informed decisions relating to communication practices and engagement strategies with students.

2.3.5 Limited Academic Workflow Orchestration

Multiple education platforms, such as Canvas, Moodle, etc. have built-in notification systems and event-driven messaging capabilities.

However, many of these systems only offer a limited number of pre-defined communication scenarios with very few options available for creating automated workflows that are unique to each institution. More complex processes like attendance-based escalation chains,

advisor intervention workflows, personalized student outreach campaigns, and context-rich academic alerts require extensive custom programming or the use of outside tools.

Additionally, most enterprise academic systems require administrative set-up and technical expertise to configure, making it very difficult for non-technical faculty or staff to access and use their systems.

The proposed SACP proposes configurable workflow orchestration that will combine automation rules, communications templates, reporting/analytics, and AI-assisted prioritization on a single, unified platform.

2.3.6 Fragmented Communication Ecosystems

A growing number of institutions have begun using two or more different solutions to communicate with their faculty members and students, through:

- Gmail
- WhatsApp
- Google Classroom
- Discord

This fragmented approach increases operational complexity, creates integration challenges and can lead to inconsistent communication experiences for both students and staff.

The primary goal of SACP is to consolidate these communication functions into a single, centralized, distributed platform to enable unified communication management.

2.3.7 Compliance, Privacy, and Institutional Governance Challenges

Institutions have a requirement to independently manage their compliance requirements associated with communication and privacy regulations, consent to communicate with students, and data governance across multiple communication platforms.

Although some systems (e.g., WhatsApp Business, TrueDialog, and enterprise email platforms) have developed security controls and consent mechanisms that provide for compliance, they lack university-based governance programs that meet the institution's communication policy objectives.

In addition, educational institutions need communication systems that can provide privacy protection, role-based access, auditing of communication, and responsible use of communications and other student information.

The SACP has been designed to create a centralized communication platform and include institutional governing mechanisms to facilitate communication and communication policy compliance.

2.3.8 Scalability and Integration Challenges

A number of reviewed systems have costly enterprise subscription requirements, are heavily reliant upon proprietary ecosystems, and offer limited customization options. Institutions

often face problems integrating communication platforms with their current academic infrastructure, creating increased implementation complexity and operational costs.

The need for scalable, easily customizable, and modularly extensible communication solution has become critical to enable educational institutions to adapt quickly and easily to changing academic needs without being overly dependent on a third-party vendor.

The proposed SACP will eliminate these challenges by providing a modularly distributed architecture that allows for extensibility, customization at the institution level, and the ability to integrate with academic systems while at the same time providing a scalable platform for future growth.